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Submitted as a phase-wise project submission.

1. Project Overview

AI BlogNest API is a comprehensive, scalable backend REST platform engineered using Node.js, Express.js, and MongoDB. It integrates Google Gemini AI to assist creators with content generation and article summarization while maintaining strict role-based platform access.

2. Implementation Summary

- **Server Framework:** Node.js & Express.js backend server utilizing modular routing, CORS, logging, and centralized error handling middleware.
- **Database Management:** MongoDB database connected via Mongoose ODM, enforcing strict schemas for Users and Blog Posts.
- **AI Services Layer:** Google Gemini Generative AI SDK (`@google/generative-ai`) integration for automated blog generation and text summarization.
- **Security Layer:** Stateless JWT authorization, password hashing with bcryptjs, and Role-Based Access Control (RBAC).

3. Key Evidence

- Fully functional REST API backend verified via HTTP testing tools (Thunder Client / Postman).
- Validated operational endpoints for Authentication, Blog CRUD operations, and Gemini AI features.

4. Security Note

All sensitive configuration data—including MongoDB Connection URIs, JWT Secret keys, and Gemini API keys—are securely maintained in isolated `.env` files and excluded from public version control via `.gitignore`.

5. Conclusion

The AI BlogNest API successfully modernizes blog content management by pairing strong architectural security standards with Generative AI capabilities, significantly lowering content creation barriers for developers and blog platforms.